

**KARNATAKA BOARD
MATHEMATICS
Class 10
2019**

Time: 3hrs

M.M: 80

General Instructions to the Candidate:

1. This Question Paper consists of 40 objective and subjective types of questions.
2. This question paper has been sealed by reverse jacket. You have to cut on the right side to open the paper at the time of commencement of the examination. Check whether all the pages of the question paper are intact.
3. Follow the instructions given against both the objective and subjective types of questions.
4. Figures in the right hand margin indicate maximum marks for the questions.
5. The maximum time to answer the paper is given at the top of the question paper. It includes 15 minutes for reading the question paper.

- I. Four alternatives are given for each of the following questions / incomplete statements. Only one of them is correct or most appropriate. Choose the correct alternative and write the complete answer along with its letter of alphabet. [8 × 1 = 8]**

1. If the n -th term of an arithmetic progression $a_n = 24 - 3n$, then its 2nd term is

- (A) 18
- (B) 15
- (C) 0
- (D) 2

Ans: 18

2. The lines represented by $2x + 3y - 9 = 0$ and $4x + 6y - 18 = 0$ are

- (A) Intersecting lines
- (B) Perpendicular lines to each other
- (C) Parallel lines
- (D) Coincident lines

Ans: Coincident lines

3. A straight line which passes through two points on a circle is

- (A) A chord
- (B) A secant
- (C) A tangent
- (D) The radius

Ans: A secant

4. If the area of a circle is 49π sq. units then its perimeter is

- (A) 7π units
- (B) 9π units
- (C) 14π units
- (D) 49π units

Ans: 14π

5. “The product of two consecutive positive integers is 30.” This can be expressed algebraically as:

- (A) $x(x+2) = 30$
- (B) $x(x-2) = 30$
- (C) $x(x-3) = 30$
- (D) $x(x+1) = 30$

Ans: $x(x+1) = 30$

6. If a and b are any two positive integers then $\text{HCF}(a, b) \times \text{LCM}(a, b)$ is equal to

- (A) $a + b$
- (B) $a - b$
- (C) $a \times b$
- (D) $a \div b$

Ans: $a \times b$

7. The value of $\cos 48^\circ - \sin 42^\circ$ is

- (A) 0
- (B) $\frac{1}{4}$
- (C) $\frac{1}{2}$
- (D) 1

Ans: 0

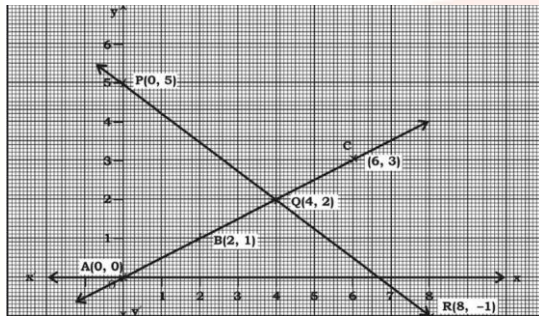
8. If $P(A) = 0.05$ then $P(\bar{A})$ is:

- (A) 0.59
- (B) 0.95
- (C) 1
- (D) 1.05

Ans: 0.95

II. Answer the following: [$6 \times 1 = 6$]

9. The given graph represents a pair of linear equations in two variables. Write how many solutions these pair of equations have.



Ans: Both the lines intersect at point Q, therefore the lines have only one pair of solution that is (4, 2).

10. $17 = 6 \times 2 + 5$ is compared with Euclid's Division lemma $a = bq + r$, then which number is representing the remainder?

Ans: In Euclid's Division lemma $a = bq + r$, r stands for remainder. Therefore, in $17 = 6 \times 2 + 5$ the remainder is 5.

11. Find the zeroes of the polynomial $P(x) = x^2 - 3$.

Ans: $P(x) = 0$

$$x^2 - 3 = 0$$

$$(x)^2 - (\sqrt{3})^2 = 0$$

$$(x + \sqrt{3})(x - \sqrt{3}) = 0$$

$$x = \pm\sqrt{3}$$

The zeroes of the polynomial $P(x) = x^2 - 3$ is $x = \pm\sqrt{3}$

12. Write the degree of the polynomial $P(x) = 2x^2 - x^3 + 5$.

Ans: The degree of the polynomial is the highest of the exponents of its various terms, therefore the degree of $P(x) = 2x^2 - x^3 + 5$ is 3.

13. Find the value of the discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$.

Ans: Discriminant (d) = $b^2 - 4ac$

In the given equation: $a = 2$; $b = -4$; $c = 3$

$$d = (-4)^2 - (4 \times 2 \times 3)$$

$$d = 16 - 24$$

$$d = -8$$

14. Write the formula to calculate the curved surface area of the frustum of a cone.

Ans: The formula to calculate the curved surface area of the frustum of a cone is $\pi l(r_1 + r_2)$

III. Answer the following:

15. Find the sum of first twenty terms of Arithmetic series $2 + 7 + 12 + \dots$ using suitable formula. [2]

Ans: $S_n = \frac{n}{2}[2a + (n-1)d]$

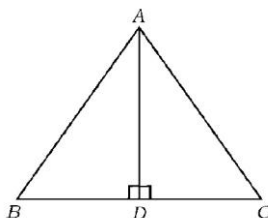
$$S_{20} = \frac{20}{2}[2(2) + (20-1)5]$$

$$S_{20} = 10[4 + 95]$$

$$S_{20} = 10(99)$$

$$S_{20} = 990$$

16. In $\triangle ABC$, $AD \perp BC$ and $AD^2 = BD \times CD$. Prove that: $AB^2 + AC^2 = (BD + CD)^2$. [2]



Ans: In the given figure, we have two right angled triangles $\triangle ADC$ and $\triangle ADB$

In $\triangle ADC$

$$AC^2 = AD^2 + DC^2 \dots\dots\dots(1)$$

In $\triangle ADB$

$$AB^2 = AD^2 + BD^2 \dots\dots\dots(2)$$

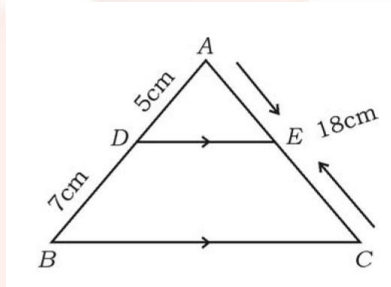
Adding (1) and (2)

$$AC^2 + AB^2 = AD^2 + DC^2 + AD^2 + BD^2$$

$$AC^2 + AB^2 = DC^2 + 2AD^2 + BD^2 \quad (\text{Given } AD^2 = BD \times CD)$$

$$AC^2 + AB^2 = (BD + CD)^2$$

17. In $\triangle ABC$, $DE \parallel BC$. If $AD = 5$ cm, $BD = 7$ cm and $AC = 18$ cm, find the length of AE .



Ans: Let $AE = x$

$$\frac{AD}{BD} = \frac{AE}{CE}$$

$$\frac{5}{7} = \frac{x}{18 - x}$$

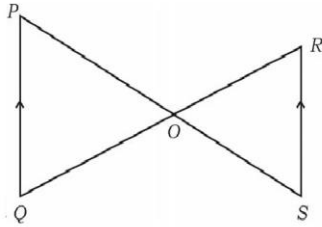
$$90 - 5x = 7x$$

$$12x = 90$$

$$x = \frac{90}{12} = 7.5$$

OR

In the given figure if $PQ \parallel RS$, prove that $\Delta POQ \sim \Delta SOR$.



Ans: In ΔPOQ and ΔSOR

$$\angle POQ = \angle SOR$$

$$\angle P = \angle S \quad (\text{Alternate Angles})$$

$$\angle Q = \angle R \quad (\text{Alternate Angles})$$

$$\therefore \Delta POQ \sim \Delta SOR \quad (\text{AAA})$$

18. Solve the following pair of linear equations by any suitable method: [2]

$$x + y = 5$$

$$2x - 3y = 5.$$

Ans:

$$x + y = 5$$

$$x = 5 - y$$

Substituting the value of x in $2x - 3y = 5$

$$2x - 3y = 5$$

$$2(5 - y) - 3y = 5$$

$$10 - 2y - 3y = 5$$

$$5 = 5y$$

$$y = 1$$

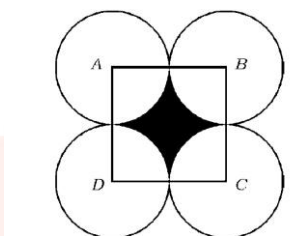
Substituting the value of y in $x = 5 - y$

$$x = 5 - y$$

$$x = 5 - 1$$

$$x = 4$$

19. In the figure, $ABCD$ is a square of side 14 cm. A , B , C and D are the centres of four congruent circles such that each circle touches externally two of the remaining three circles. Find the area of the shaded region. [2]



Ans:

Area of shaded region = Area of Square $ABCD$ – Area of 4 quadrants of circle

Side of square = 14cm

Area of Square = $(side)^2 = (14)^2 = 256cm^2$

Radius of circle = $\frac{14}{2} = 7cm$

Area of quadrant of circle = $\frac{1}{4}\pi r^2$

Area of 4 quadrants of circle = $4\left(\frac{1}{4}\pi r^2\right) = \frac{22}{7} \times 7 \times 7 = 154cm^2$

Area of shaded region = $256 - 154 = 102cm^2$

20. Draw a circle of radius 4 cm and construct a pair of tangents such that the angle between them is 60° . [2]

Ans: The steps of construction are:

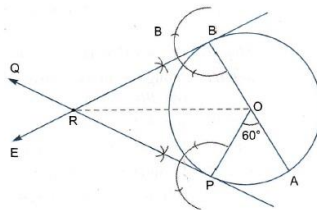
Step – 1: Draw a circle with centre O and radius 4cm.

Step – 2: Draw any diameter AOB .

Step – 3: Make $\angle AOP = 60^\circ$

Step – 4: Draw $PQ \perp OP$ and $BE \perp OB$. PQ and BE intersect at R .

Step – 5: Hence, RB and RP are the required tangents.



21. Find the co-ordinates of point which divides the line segment joining the points A (4, -3) and B (8, 5) in the ratio 3: 1 internally. [2]

Ans: Using Section Formula

$$P = \left(\frac{x_2m + x_1n}{m+n}, \frac{y_2m + y_1n}{m+n} \right)$$

Here $x_1 = 4$, $y_1 = -3$, $x_2 = 8$, $y_2 = 5$, $m = 3$, $n = 1$

$$P = \left(\frac{8(3) + 4(1)}{3+1}, \frac{5(3) + (-3)1}{3+1} \right)$$

$$P = \left(\frac{24+4}{4}, \frac{15-3}{4} \right)$$

$$P = \left(\frac{28}{4}, \frac{12}{4} \right)$$

$$P = (7, 3)$$

22. Prove that $3 + \sqrt{5}$ is an irrational number. [2]

Ans: Let's assume $3 + \sqrt{5}$ is a rational number

$$3 + \sqrt{5} = \frac{a}{b}$$

$$\sqrt{5} = \frac{a}{b} - 3$$

$$\sqrt{5} = \frac{a}{b} - \frac{3b}{b}$$

$$\sqrt{5} = \frac{a-3b}{b}$$

As we know that $\sqrt{5}$ is an irrational number so, $\frac{a-3b}{b}$ is also an irrational number. Therefore, our assumption is wrong.

Hence, $3 + \sqrt{5}$ is an irrational number.

23. The sum and product of the zeroes of a quadratic polynomial $P(x) = ax^2 + bx + c$ are -3 and 2 respectively. Show that $b + c = 5a$. [2]

Ans: In $P(x) = ax^2 + bx + c$

$$\text{Sum} = \frac{-b}{a} = -3 \Rightarrow b = 3a$$

$$\text{Product} = \frac{c}{a} = 2 \Rightarrow c = 2a$$

$$b + c = 3a + 2a$$

$$b + c = 5a$$

24. Find the quotient and the remainder when $P(x) = 3x^3 + x^2 + 2x + 5$ is divided by $g(x) = x^2 + 2x + 1$. [2]

Ans: The quotient and remainder when $P(x) = 3x^3 + x^2 + 2x + 5$ is divided by $g(x) = x^2 + 2x + 1$ are $3x - 5$ and $9x + 10$ respectively.

$$\begin{array}{r} \overline{3x - 5} \\ x^2 + 2x + 1 \overline{) 3x^3 + x^2 + 2x + 5} \\ \underline{-(3x^3 + 6x^2 + 3x)} \\ -5x^2 - x + 5 \\ \underline{-(-5x^2 - 10x - 5)} \\ 9x + 10 \end{array}$$

25. Solve $2x^2 - 5x + 3 = 0$ by using formula. [2]

$$\text{Ans: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Where $a = 2; b = -5; c = 3$

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(2)(3)}}{2(2)}$$

$$x = \frac{5 \pm \sqrt{25 - 24}}{4}$$

$$x = \frac{5 \pm \sqrt{1}}{4}$$

$$x_1 = \frac{5+1}{4}; x_2 = \frac{5-1}{4}$$

$$x_1 = 1.5; x_2 = 1$$

26. The length of a rectangular field is 3 times its breadth. If the area of the field is 147 sq. m, find its length and breadth. [2]

Ans: Let the breadth of the field be “x”, then the length will be “3x”

Area of field = length X breadth

$$147 = 3x(x)$$

$$147 = 3x^2$$

$$x^2 = \frac{147}{3}$$

$$x^2 = 49$$

$$x = 7$$

Therefore, the breadth of the rectangular field is 7m and length is 21m.

27. If $\sin \theta = \frac{12}{13}$, find the values of $\cos \theta$ and $\tan \theta$. [2]

Ans: Given $\sin \theta = \frac{12}{13}$

$$\sin \theta = \frac{P}{H} = \frac{12}{13}$$

$$\Rightarrow P = 12 \text{ and } H = 13$$

$$H^2 = P^2 + B^2$$

$$169 = 144 + B^2$$

$$169 - 144 = B^2$$

$$25 = B^2$$

$$B = 5$$

$$\cos \theta = \frac{B}{H} = \frac{5}{13}$$

$$\tan \theta = \frac{P}{B} = \frac{12}{5}$$

OR

If $\sqrt{3}\tan \theta = 1$ and θ is acute, find the value of $\sin 3\theta + \cos 2\theta$.

Ans:

$$\sqrt{3}\tan \theta = 1$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta = 30^\circ$$

$$\sin 3\theta + \cos 2\theta$$

$$= \sin(3 \times 30) + \cos(2 \times 30)$$

$$= \sin 90^\circ + \cos 60^\circ$$

$$= 1 + \frac{1}{2} = \frac{3}{2}$$

28. Prove that $\left(\frac{1 + \cos \theta}{1 - \cos \theta} \right) = (\operatorname{cosec} \theta + \cot \theta)^2$. [2]

Ans: $\left(\frac{1 + \cos \theta}{1 - \cos \theta} \right) = (\operatorname{cosec} \theta + \cot \theta)^2$

Taking R.H.S

$$\begin{aligned}
 & (\operatorname{Cosec} \theta + \cot \theta)^2 \\
 &= \operatorname{Cosec}^2 \theta + 2 \operatorname{Cosec} \theta \cot \theta + \cot^2 \theta \\
 &= \frac{1}{\sin^2 \theta} + 2 \times \frac{1}{\sin \theta} \times \frac{\cos \theta}{\sin \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} \\
 &= \frac{1}{\sin^2 \theta} + \frac{2 \cos \theta}{\sin^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} \\
 &= \frac{1 + 2 \cos \theta + \cos^2 \theta}{\sin^2 \theta} \\
 &= \frac{(1 + \cos \theta)^2}{1 - \cos^2 \theta} \\
 &= \frac{(1 + \cos \theta)^2}{(1 + \cos \theta)(1 - \cos \theta)} \\
 &= \frac{1 + \cos \theta}{1 - \cos \theta}
 \end{aligned}$$

Hence L.H.S = R.H.S

29. A cubical die numbered from 1 to 6 is rolled twice. Find the probability of getting the sum of numbers on its faces is 10. [2]

Ans: If a dice is rolled twice, then the sample space $n(s)$ will be 36

Let A be the event of getting sum of numbers on its faces 10, then $n(A) = (4, 6) (5, 5), (6, 4) = 3$

Therefore, the probability of getting sum of numbers on its faces 10 = $\frac{3}{36} = \frac{1}{12}$

30. The radii of two circular ends of a frustum of a cone shaped dustbin are 15 cm and 8 cm. If its depth is 63 cm, find the volume of the dustbin. [2]

Ans: Volume = $\frac{\pi}{3}(r_1^2 + r_1 r_2 + r_2^2)h$

Where

$$r_1 = 15$$

$$r_2 = 8$$

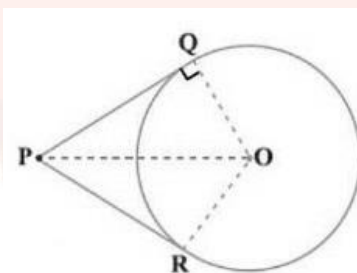
$$h = 63$$

$$\text{Volume} = \frac{\pi}{3}(225 + 120 + 64)63 = \frac{22}{21} \times 409 \times 63 = 22 \times 409 \times 3 = 26994 \text{ cu.cm}$$

IV. Answer the following:

31. Prove that “the lengths of tangents drawn from an external point to a circle are equal”. [3]

Ans: Let the circle be with centre O and P be a point outside the circle. PQ and PR are two tangents to circle intersecting at point Q and R respectively as shown in the figure below.



As PQ is a tangent so $OQ \perp PQ$ (Tangent at any point of circle is perpendicular to the radius through the point of contact.)

So $\angle OQP = 90^\circ$

Hence $\triangle OQP$ is right triangle.

Similarly, $\triangle ORP$ is also right triangle.

In $\triangle OQP$ and $\triangle ORP$

$$\angle OQP = \angle ORP = 90^\circ$$

$OP = OP$ (Common side)

$OQ = OR$ (radius)

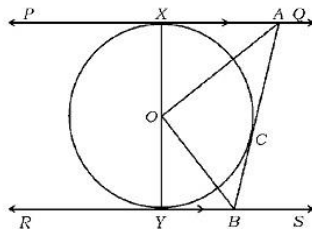
$\triangle OQP = \triangle ORP$ (R.H.S Congruency)

Hence, $PQ = PR$ (Corresponding parts of Congruent Triangles)

Hence Proved.

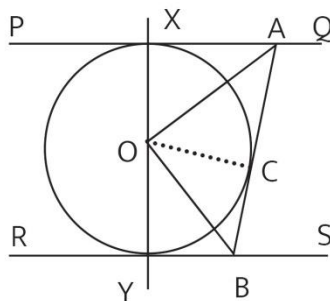
OR

In the given figure PQ and RS are two parallel tangents to a circle with centre O and another tangent AB with point of contact C intersecting PQ at A and RS at B . Prove that $\angle AOB = 90^\circ$.



Ans:

Join OC



In $\triangle OXA$ and $\triangle OCA$

$OX = OC$ (radius)

$AX = AC$ (tangents drawn from point A)

$AO = AO$ (common sides)

Therefore, $\triangle OXA \cong \triangle OCA$ (SSS)

Hence, $\angle XOA = \angle COA$(1)

Similarly in $\triangle OYB$ and $\triangle OCB$

$\angle YOB = \angle COB$(2)

XOY is the diameter of the circle, therefore $\angle XOA + \angle COA + \angle YOB + \angle COB = 180^\circ$

From (1) and (2)

$$2\angle COA + 2\angle COB = 180^\circ$$

$$\angle COA + \angle COB = 90^\circ$$

$$\angle AOB = 90^\circ$$

Hence proved.

32. Calculate the median of the following frequency distribution table: [3]

Class Interval	Frequency
1 – 4	6
4 – 7	30
7 – 10	40
10 – 13	16
13 – 16	4
16 – 19	4

Ans:

Class Interval	Frequency	Cumulative Frequency
1 – 4	6	6
4 – 7	30	36
7 – 10	40	76
10 – 13	16	92
13 – 16	4	96
16 – 19	4	100

Median Class = Size of $\left(\frac{N}{2}\right)$ th class = Size of 50th class = 7 – 10

$$\text{Median} = L + \left(\frac{\frac{N}{2} - cf}{f} \times i \right) = 7 + \left(\frac{50 - 36}{40} \times 3 \right) = 7 + \frac{21}{20} = \frac{161}{20} = 8.05$$

OR

Calculate the mode for the following frequency distribution table.

Class Interval	Frequency
10 – 25	2
25 – 40	3
40 – 55	7
55 – 70	6
70 – 85	6
85 – 100	6

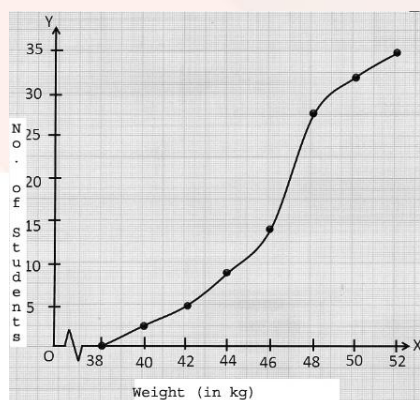
Ans: The class having highest frequency is 40 – 55, therefore, it is the model class

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i \right) = 40 + \left(\frac{7 - 3}{14 - 3 - 6} \times 15 \right) = 40 + \frac{60}{5} = 40 + 12 = 52$$

33. During the medical check-up of 35 students of a class, their weights were recorded as follows. Draw a less than type of ogive for the given data: [3]

Class Interval	Frequency
Less than 38	0
Less than 40	3
Less than 42	5
Less than 44	9
Less than 46	14
Less than 48	28
Less than 50	32
Less than 52	35

Ans:



34. The seventh term of an Arithmetic progression is four times its second term and twelfth term is 2 more than three times of its fourth term. Find the progression. [3]

Ans: The seventh term of an AP = $a_7 = 4(a_2)$

The twelfth term of the AP = $a_{12} = 2 + 3(a_4)$

$$a_7 = 4(a_2)$$

$$a + 6d = 4(a + d)$$

$$a + 6d = 4a + 4d$$

$$2d = 3a \dots\dots\dots(1)$$

$$a_{12} = 2 + 3(a_4)$$

$$a + 11d = 2 + 3(a + 3d)$$

$$a + 11d = 2 + 3a + 9d$$

$$2d = 2a + 2 \dots\dots\dots(2)$$

From (1) and (2)

$$3a = 2a + 2$$

$$a = 2$$

$$d = 3$$

Therefore, the progression is: 2, 5, 8,

OR

A line segment is divided into four parts forming an Arithmetic progression. The sum of the lengths of 3rd and 4th parts is three times the sum of the lengths of first two parts. If the length of fourth part is 14 cm, find the total length of the line segment.

Ans: Given

$$a_3 + a_4 = 3(a_1 + a_2)$$

$$a_4 = 14$$

$$(a + 2d) + 14 = 3(a + a + d)$$

$$a + 2d + 14 = 6a + 3d$$

$$14 = 5a + d \dots\dots\dots(1)$$

$$a_4 = 14$$

$$a + 3d = 14 \dots\dots\dots(2)$$

From (1) and (2)

$$a + 3d = 5a + d$$

$$2d = 4a$$

$$d = 2a$$

From (2)

$$a + 3d = 14$$

$$a + 3(2a) = 14$$

$$a + 6a = 14$$

$$7a = 14$$

$$a = 2$$

$$d = 4$$

$$a_1 = 2$$

$$a_2 = 6$$

$$a_3 = 10$$

$$a_4 = 14$$

Therefore, length of the line segment is $2 + 6 + 10 + 14 = 32cm$

35. The vertices of a ΔABC are $A(-3, 2)$, $B(-1, -4)$ and $C(5, 2)$. If M and N are the midpoints of AB and AC respectively, show that $2MN = BC$. [3]

Ans: Given: $A(-3, 2)$, $B(-1, -4)$ and $C(5, 2)$ and M is the midpoint of AB and N is the midpoint of AC .

Using the formula of midpoint: $x = \frac{x_1 + x_2}{2}$; $y = \frac{y_1 + y_2}{2}$

$$\text{For Point M: } x = \frac{-3-1}{2} = -2; y = \frac{2-4}{2} = -1 = (-2, -1)$$

$$\text{For Point N: } x = \frac{-3+5}{2} = 1; y = \frac{2+2}{2} = 2 = (1, 2)$$

Using distance Formula: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$\text{Distance between M and N} = \sqrt{(1+2)^2 + (2+1)^2}$$

$$MN = \sqrt{3^2 + 3^2}$$

$$= \sqrt{9+9}$$

$$= \sqrt{18}$$

$$= 3\sqrt{2}$$

$$\text{Distance between B and C} = \sqrt{(5+1)^2 + (2+4)^2}$$

$$\begin{aligned} MN &= \sqrt{6^2 + 6^2} \\ &= \sqrt{36 + 36} \\ &= \sqrt{72} \\ &= 6\sqrt{2} \end{aligned}$$

Therefore, $BC = 2MN$

OR

The vertices of a ΔABC are $A(-5, -1)$, $B(3, -5)$, $C(5, 2)$. Show that the area of the ΔABC is four times the area of the triangle formed by joining the mid-points of the sides of the triangle ABC .

$$\text{Ans: Area of Triangle} = \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$$

$$\text{Coordinates of midpoints: } x = \frac{x_1 + x_2}{2}; y = \frac{y_1 + y_2}{2}$$

Let the midpoints of AB , BC and AC be P , Q and R respectively.

$$\text{For Point P: } x = \frac{-5+3}{2} = -1; y = \frac{-1-5}{2} = -3$$

$$\text{For Point Q: } x = \frac{3+5}{2} = 4; y = \frac{-5+2}{2} = \frac{-3}{2} = -1.5$$

$$\text{For Point R: } x = \frac{-5+5}{2} = 0; y = \frac{-1+2}{2} = \frac{1}{2} = 0.5$$

$$\text{Area of Triangle } ABC = \frac{1}{2} [-5(-5-2) + 3(2+1) + 5(-1+5)] = \frac{1}{2} [-5(-7) + 3(3) + 5(4)] = 32$$

Area of Triangle PQR

$$= \frac{1}{2} [-1(-1.5-0.5) + 4(0.5+3) + 0(-3+1.5)] = \frac{1}{2} [-1(-2) + 4(3.5) + 0(-1.5)] = 8$$

Therefore, Area of Triangle $ABC = 4$ (Area of Triangle PQR)

36. Construct a triangle with sides 5 cm, 6 cm and 7 cm and then construct another triangle whose sides are $\frac{7}{5}$ of the corresponding sides of the first triangle. [3]

Ans: At first construct a triangle ABC for the given measurements:

1. Construct a line segment AB of 7cm with help of scale.
2. Then with help of compass put an arc at a distance of 5cm with A as center and another arc at a distance of 6cm with B as centre.
3. Name the point as C where both the arcs cut.
4. Join AC and BC.

Now for another triangle:

1. Draw a ray AX making an acute angle with AB on the side opposite to the vertex C.
2. Mark 7 points on the ray naming as A1 to A7. Join B and A5.
3. Extend the line AB and name it AB' and join B' with A7.
4. Now draw a line parallel to BC from B' and name it B'C'.
5. The triangle AB'C' is the required triangle.

V. Answer the following:

37. Find the solution of the following pairs of linear equation by the graphical method: 4

$$2x + y = 6$$

$$2x - y = 2$$

Ans:

$$2x + y = 6$$

$$y = 6 - 2x$$

Now we will find the values of y for different values of x

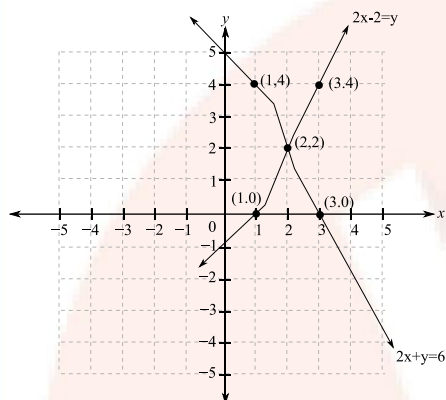
x	1	2	3
y	4	2	0

$$2x - y = 2$$

$$y = 2x - 2$$

Now we will find the values of y for different values of x

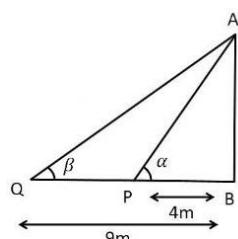
x	1	2	3
y	0	2	4



Solution point is (2, 2). Therefore, $x = 2$ and $y = 2$

38. The angles of elevation of the top of a tower from two points at a distance of 4 m and 9 m from the base of the tower and in the same straight line with it are complementary. Find the height of the tower. [4]

Ans: From the above information we can get



It is given that α and β are supplementary so, $\alpha + \beta = 90^\circ$

In $\triangle ABP$

$$\tan \alpha = \frac{AB}{BP} \dots\dots\dots(1)$$

In $\triangle ABQ$

$$\tan \beta = \frac{AB}{BQ}$$

$$\tan (90 - \alpha) = \frac{AB}{BQ}$$

$$\cot \alpha = \frac{AB}{BQ}$$

$$\tan \alpha = \frac{BQ}{AB} \dots\dots\dots(2)$$

From (1) and (2)

$$\frac{BQ}{AB} = \frac{AB}{BP}$$

$$BQ \times BP = (AB)^2$$

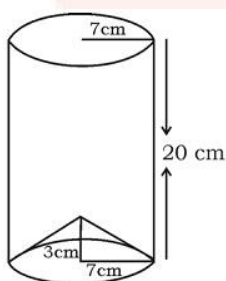
$$9 \times 4 = (AB)^2$$

$$(AB)^2 = 36$$

$$AB = 6m$$

Therefore, height of the tower is 6m.

39. The bottom of a right cylindrical shaped vessel made from metallic sheet is closed by a cone shaped vessel as shown in the figure. The radius of the circular base of the cylinder and radius of the circular base of the cone are each is equal to 7 cm. If the height of the cylinder is 20 cm and height of cone is 3 cm, calculate the cost of milk to fill completely this vessel at the rate of Rs. 20 per litre.



Ans:

$$\text{Volume of cylinder} = \frac{1}{3} \pi r^2 h$$

$$= \frac{22}{7} \times 7 \times 7 \times 20$$

$$= 3080 \text{ cm}^3$$

$$\text{Volume of Cone} = \pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times 3$$

$$= 154 \text{ cm}^3$$

$$\text{Total volume of vessel} = 3080 - 154 = 2926 \text{ cm}^3$$

$$\text{The cost of filling milk} = 2926 \text{ cm}^3 \times 20 = \text{Rs.} 58520$$

OR

A hemispherical vessel of radius 14 cm is fully filled with sand. This sand is poured on a level ground. The heap of sand forms a cone shape of height 7 cm. Calculate the area of ground occupied by the circular base of the heap of the sand.

Ans:

$$\text{Volume of Hemisphere} = \text{Volume of heap of Sand}$$

$$\frac{2}{3} \pi r^3 = \frac{1}{3} \pi r^2 h$$

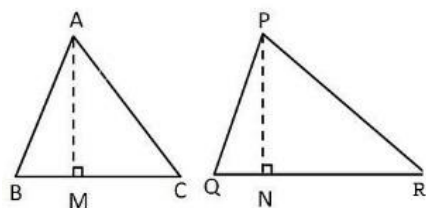
$$2 \times 2744 = 7r^2$$

$$r^2 = 784$$

$$r = 28$$

40. Prove that “the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides”. [4]

Ans: Let the two triangles be:



$$\text{Area of } \triangle ABC = \frac{1}{2} \times BC \times AM \dots\dots\dots(1)$$

$$\text{Area of } \triangle PQR = \frac{1}{2} \times QR \times AN \dots\dots\dots(2)$$

Dividing (1) by (2)

$$\frac{\text{ar}(ABC)}{\text{ar}(PQR)} = \frac{\frac{1}{2} \times BC \times AM}{\frac{1}{2} \times QR \times PN}$$

$$\frac{\text{ar}(ABC)}{\text{ar}(PQR)} = \frac{BC \times AM}{QR \times PN} \dots\dots\dots(1)$$

In $\triangle ABM$ and $\triangle PQN$

$\angle B = \angle Q$ (Angles of similar triangles)

$\angle M = \angle N$ (Both 90°)

Therefore, $\triangle ABM \sim \triangle PQN$

$$\text{So, } \frac{AB}{AM} = \frac{PQ}{PN} \dots\dots\dots(2)$$

From 1 and 2

$$\frac{\text{ar}(ABC)}{\text{ar}(PQR)} = \frac{BC}{QR} \times \frac{AM}{PN}$$

$$\Rightarrow \frac{\text{ar}(ABC)}{\text{ar}(PQR)} = \frac{BC}{QR} \times \frac{AB}{PQ} \dots\dots\dots(3)$$

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} \dots\dots\dots(\triangle ABC \sim \triangle PQR)$$

Putting in (3)

$$\frac{\text{ar}(ABC)}{\text{ar}(PQR)} = \frac{AB}{PQ} \times \frac{AB}{PQ} = \left(\frac{AB}{PQ}\right)^2$$

$$\Rightarrow \frac{\text{ar}(ABC)}{\text{ar}(PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{BC}{QR}\right)^2 = \left(\frac{AC}{PR}\right)^2$$